MathQA Benchmark Report

AgentOpt Model Selection Framework

9 Bedrock Models | 200 Samples | 2-Tuple (Answer+Critic) | March 2026

1. Brute Force Ground Truth (Answer + Critic)

- Exhaustive evaluation: all 81 combos (9x9) x 200 MathQA samples.
- 2-tuple architecture: answer model solves math, critic model reviews and corrects.
- Best combo: Opus + Qwen3 Next at 98.83% accuracy.
- Total brute force cost across all combos: $113.01.
- Total evaluations: 14,855 of 16,200 (8.3% skipped due to API errors).
  Error rate varies by answer model: Ministral 3 8B (19.9%), C3 Haiku (18.8%),
  Opus (14.5%), Haiku 4.5 (0.1%). The self-reflective loop (up to 3 rounds) generates
  more API calls per sample, increasing failure probability under 81-combo concurrency.
  Failures are API-level (timeouts/throttling), likely random w.r.t. difficulty.
  Accuracy is computed over successfully completed samples only (failures excluded
  from denominator, not scored as 0). This avoids penalizing models for infrastructure
  issues but means combos with high error rates have fewer effective samples.

Top 15 Combinations (of 81)

Rank	Answer + Critic	Accuracy	Cost
1	**Claude Opus 4.6 + Qwen3 Next 80B A3B**	**98.83%**	$5.89
2	Claude Opus 4.6 + Ministral 3 8B	98.73%	$5.31
3	Claude Opus 4.6 + Claude Haiku 4.5	98.27%	$6.09
4	Claude Opus 4.6 + Qwen3 32B	97.79%	$6.42
5	Claude Opus 4.6 + Claude Opus 4.6	97.77%	$6.95
6	Claude Opus 4.6 + gpt-oss-120b	97.73%	$6.14
7	Claude Opus 4.6 + Claude 3 Haiku	97.26%	$5.26
8	Claude Opus 4.6 + Kimi K2.5	97.25%	$6.66
9	Claude Opus 4.6 + gpt-oss-20b	97.13%	$6.10
10	Claude Haiku 4.5 + Ministral 3 8B	94.47%	$2.59
11	Claude Haiku 4.5 + Claude Haiku 4.5	94.00%	$3.17
12	Claude Haiku 4.5 + Claude Opus 4.6	94.00%	$3.89
13	Claude Haiku 4.5 + Qwen3 Next 80B A3B	94.00%	$2.50
14	Ministral 3 8B + Claude 3 Haiku	93.98%	$0.05
15	Claude Haiku 4.5 + Kimi K2.5	93.97%	$2.92

Bottom 10 Combinations (of 81)

Rank	Answer + Critic	Accuracy	Cost
72	gpt-oss-120b + gpt-oss-20b	77.32%	$0.17
73	Kimi K2.5 + Claude 3 Haiku	76.96%	$0.92
74	Qwen3 Next 80B A3B + Kimi K2.5	76.96%	$0.46
75	Qwen3 Next 80B A3B + gpt-oss-120b	76.84%	$0.32

76	gpt-oss-120b + Qwen3 Next 80B A3B	74.74%	$0.20
77	Claude 3 Haiku + gpt-oss-20b	72.96%	$0.31
78	Kimi K2.5 + Qwen3 32B	72.77%	$0.67
79	Claude 3 Haiku + Qwen3 Next 80B A3B	68.94%	$0.36
80	Claude 3 Haiku + Qwen3 32B	63.86%	$0.27
81	Claude 3 Haiku + Claude 3 Haiku	59.88%	$0.30

Note: Per-sample latency not reported. 81 concurrent combos cause API-side queueing that inflates per-sample times (same issue as HotpotQA). Only GPQA latencies are reliable.

2. Offline Selector Simulation (50 Seeds)

- Per-sample scores from brute force replayed with each selector's logic
  -- zero additional API calls required.
- Each selector simulated 50 times with different random seeds.
- Hill climbing uses topology-aware neighbors (quality/speed ranking).
  HC (1) = 1 random restart; HC (3) = 3 restarts (library default).
- Metrics: Find Rate = % finding true best. Mean Acc = avg accuracy of chosen combo.

Selector	Found Best	Mean Acc	Evaluations	Cost	Savings
Brute Force	**100%**	**98.83%**	**14,855**	**$113.01**	**--**
Arm Elimination	**96%**	**98.80%**	**3,632**	**$61.22**	**46%**
Random Search	28%	98.04%	3,850	$28.83	74%
Hill Climbing (3)	72%	97.81%	4,058	$45.72	60%
Epsilon LUCB	0%	97.46%	443	$5.55	95%
LM Proposal	0%	96.87%	149	$5.15	95%
Bayesian Opt.	4%	95.39%	3,608	$31.05	73%
Hill Climbing (1)	28%	93.53%	1,610	$18.26	84%
Threshold SE	0%	77.23%	369	$1.95	98%

3. Key Takeaways

Brute Force Insights

- Opus as answerer dominates: ALL 9 Opus combos are ranked 1-9 (97-99%).
  Critic choice barely matters -- only ~1.7% spread across all critics.
  Opus solves math correctly on the first attempt; the critic rarely overrides.
- Best value: Opus + Ministral (98.73%) -- cheapest critic, near-identical accuracy.
  Cost: $5.31 vs $6.95 for Opus+Opus. Expensive critic adds cost, not quality.
- Surprising: Ministral + C3 Haiku (93.98%) at $0.05 -- 113x cheaper than #1 ($5.89)
  with only a 5% accuracy drop. The cheapest models can be remarkably effective.
- gpt-oss-20b + gpt-oss-120b achieves 93.96% at $0.12 -- strong budget option.
- Clear answer-model tiers: Opus (97-99%) > Haiku 4.5 (91-94%) >
  gpt-oss-20b/Ministral (88-94%) > Qwen3 32B (82-89%) > others (60-82%).
- Ranking shift from 40-sample pilot: Haiku 4.5 was best answerer at 97.5% (ceiling
  effect with only 40 samples). At 200 samples, Opus pulls ahead (98.83% vs 94%).
  The 40-sample run was too noisy to distinguish saturated models.
- Contrast with HotpotQA: there, the strongest model (Opus) fails as planner.
  Here, Opus excels as answerer. The optimal role depends on the task architecture.

Selector Insights

- Arm Elimination wins again: 96% find rate, consistent across all 4 benchmarks.
- Hill Climbing (3) is useful for 2-tuple: 72% find rate, 60% savings.
  Topology-aware neighbors help navigate the 81-combo space.
- Epsilon LUCB and Threshold SE both get 0% find rate.
  The top 9 combos are ALL Opus-as-answerer with tiny differences (~1.7% spread).
  Bandits cannot distinguish near-identical arms with limited samples.
- Random Search has high mean accuracy (98.04%) despite low find rate (28%) --
  because ANY Opus combo is near-optimal. Hard to pick wrong when 9/81 are >97%.
- Bayesian Optimization underperforms (4%) -- discrete 2-tuple space is a poor fit
  for GP-based surrogate modeling.
- LM Proposal achieves high accuracy (96.87%) despite 0% find rate -- GPT-4.1 picks
  Opus combos but not the exact best one. 149 avg evaluations (not 200) reflects
  the varying sample counts per combo due to API errors. Like random search,
  hard to go wrong when any Opus combo scores >97%.

4. Why the Critic Barely Matters (for Strong Answerers)

- MathQA uses a self-reflective architecture: the answer model solves the problem, then the critic reviews and optionally corrects the answer.
- When Opus answers, it achieves 97-99% regardless of critic (~1.7% spread). The critic has almost nothing to correct -- Opus gets math right on the first try.
- This means the critic is essentially a no-op for strong answerers.

Opus as Answerer: Critic Impact (9 combos)

Critic	Accuracy	Cost	Delta vs Best
Qwen3 Next 80B A3B	**98.83%**	$5.89	--
Ministral 3 8B	98.73%	$5.31	-0.10%
Claude Haiku 4.5	98.27%	$6.09	-0.56%
Qwen3 32B	97.79%	$6.42	-1.04%
Claude Opus 4.6	97.77%	$6.95	-1.07%
gpt-oss-120b	97.73%	$6.14	-1.10%
Claude 3 Haiku	97.26%	$5.26	-1.57%
Kimi K2.5	97.25%	$6.66	-1.58%
gpt-oss-20b	97.13%	$6.10	-1.70%

- The spread across all 9 critics is only 1.70% (98.83% to 97.13%).
- The cheapest critic (Ministral, $5.31) performs nearly identically to the best.
- Contrast: for weak answerers (C3 Haiku), the critic matters more -- C3 Haiku + Opus (92.94%) vs C3 Haiku + C3 Haiku (59.88%) = 33% gap.
- Implication: pair strong answerers with cheap critics to save cost. Pair weak answerers with strong critics if budget allows.

Selector Cost Efficiency: Why Savings Are Moderate

- Arm Elimination saves only 46% despite finding the best combo 96% of the time.
- This is because the best combo uses Opus as answerer -- the most expensive model.
- Any selector that finds the best must evaluate Opus combos, which are costly.
- The 9 Opus-as-answerer combos alone cost ~$53 (47% of total $113).
- Selectors cannot avoid this cost if the true best is an Opus combo.
- Compare with HotpotQA: best combo uses cheap planner (Ministral), so eliminating expensive planner combos early saves 64%. Here, the expensive model is essential.

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